

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: **5**

Annexure A

ANALYTICAL PROBLEM SOLVING

Total **Marks:**50

Code of Conduct:

1B · Rajesh

(Name/Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Twigly.

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and

Admin

/28. Using the given subnet masks, apply subnetting techniques to assign

suitable subnet addresses for each department. Determine the network address,

valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created

and the number of usable hosts per subnet. Determine the subnet to which the IP

address **172.16.10.78** belongs. Identify the network address and broadcast address

for that subnet. Also, determine whether the IP address **172.16.10.95** falls within

the same subnet range based on the given configuration.

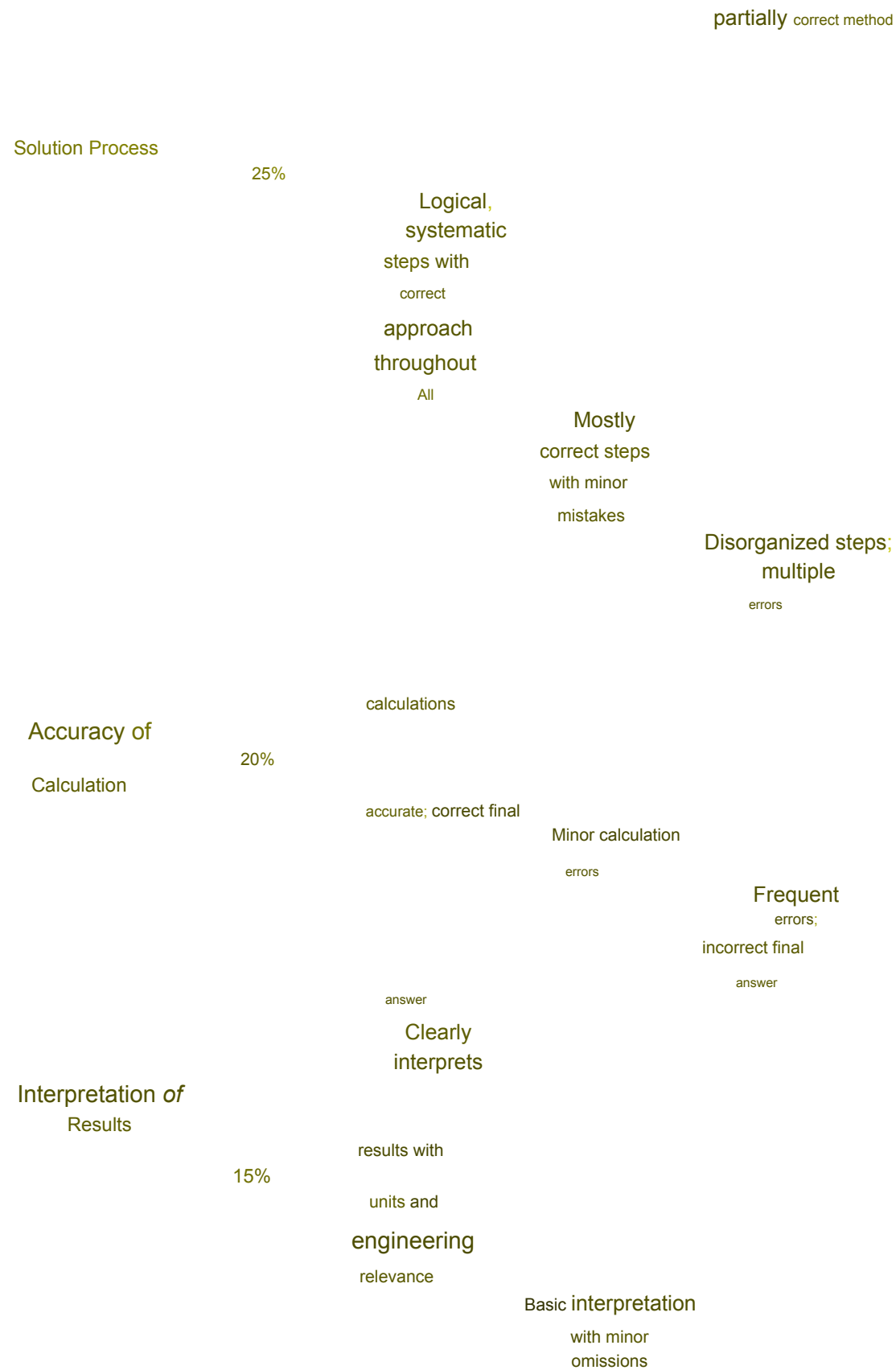
5. A network uses the Internet Protocol version 6 address

2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible

host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

		Level 4 (Exemplary)				Level 3 (Proficient)		Level 2 (Developing)		Level 1 (Beginning)	
Criteria	Weightage										
Problem Understanding	20%	Clearly interprets problem,				Understands					
		identifies all parameters				problem with minor gaps		Partial understanding; misses key elements		Misinterprets or unable to understand problem	
Method Selection	20%	and requirements accurately Selects most appropriate method/formu				Selects correct method with					
		la with strong justification				minor errors				Inappropriate or	



Limited or unclear
interpretation

Unable to
select method

No clear
process

No valid calculations

No
interpretation

2) subnetting problem-

Given!

* System | IP:

192.168.1.130

* system 2 IP:

192. 168.1.190

* subnet mask: 255.255.255.192 (126)

stept-

Find the subnet mask

126 -> BLOCK size=64

*subnets

192.168.1.0-63

* 192.168.1-64 -127

* 192.168.1.128 - 191

कै

192.168.1-192-255

Systemi !

IP Address: 192.168.1.190

Network Address: 192.168.1.128.

Broadcast Address: 192.168.1.191

host **Range:** 192.168.1.129 -
192.168.1.190

valid

IP 192.168.1.190

belongs

to this subnet:yes

usable

hosts: 62

Final

answer:

System

Ipaddres Network

Host

Broadcast

192.168.1.128

129-

192.168.1-191

System 1

192.168.1.130

190

system 2

192.168.1.190 192.168.1.128

129-

190

192.168.0.1-191

observation!

a router

Both systems belong to the same subnet

- They can communicate directly without a router because they are in the same network.

Conclusion :

The

subnet mask 255.255.255-192 (/26) creates Subnets with 62 usable host addresses . Both ip address

belong to the subnet 192.168.1.128/26, whose broadcast address is 192.168.1-191

belong

3) Given:

Network address = 172.16.0.0/16

Departments :

HR 125

- Sales -> 126

IT -> 127

Admin 128

stept: Allocates

Subnet mask!

255.255.255.128

BLOCK size:

$256 - 128 = 128$

He Department (125)

Network Address:

172.16.0.0

Host

Range !

172.16.0.1 - 172.16.0.126

wable Hosts!

27-2

$= 128 - 2$

stepa's Allocale

sales department (126) .

Next available address:

172.16.0.128

subnet mask!

255.255.255.192

Block size: 64

Network Address: 172.16.0.128

Host

Range: 172.16.0.129 - 172.16.0.190

Broadcast address: 172.16.0.191

usable

Hosts: $26 - 2 = 62$

Steps: Allocates IT Department (127)

Next available

172.16.0.192

address:

Subnet mask: 255.255.255.224

Block size: 32

Network address 172.16.0.192

Host Range! 172.16.0.193-172.16.0.222

Broadcast Address:

172.16.0.223

usable Hosts: $25 - 2 = 30$

stepus **Allocates Admin Department (128)**

Next available address:

172.16.0.224

subnet mask!

255.255.255.240

Block Size: 16

Network address: 172.16.0.224

Host Range!

172.16.0.225-172.16.0.238

Broadcast address :

172.16.0.239

usable

HOJIS:

24-2=14

**Remaining unused Address
Range:-**

Allocated addresses end at:

172.16.0.239

Remaining available address

172.16.0-240 - 172.16.255.255